PACLAND · Climate, in plain words

What we checked, what we found, and what it means. One page, no jargon.

1 · The question

Climate models predict the future. Nobody can check a prediction about the year 2100, because 2100 has not happened. So the useful question is smaller, and answerable: can we check whether a model deserves to be believed at all?

2 · What we did

We borrowed a rule from physics: don't ask a model for its answer — ask what it is not allowed to do. Physics says energy cannot be created or destroyed. So the heat a climate model takes in must turn up somewhere: the ocean, the air, the land, the ice. If it doesn't, the model is leaking. Leaking is not a matter of opinion, and it is easy to check.

3 · What we found

- Heat leaks, in most models. Of the models with the data to check it, 15 of 19 leak heat. One leaks almost twice as much heat as the amount of warming it is trying to predict. Picture a bucket with a hole: the level you predict is wrong by the size of the hole.
- The simplest guess is hard to beat. We asked whether a model beats a ruler — take the warming we have already measured and continue it in a straight line. On real data, most models lose. A ruler often beats a supercomputer, because a ruler cannot be wrong in complicated ways.
- The leak is in the join, not the ocean. Salt is conserved far better than heat. Salt only moves in water; heat crosses between the air and the sea. So the mistake is where the model connects atmosphere and ocean — which is good news, because that tells you what to fix.

4 · How we score a model

For each model we count what it was actually checked against: how warm it says the world will get, how strongly it traps heat, and whether it leaks. The best of our three models scores three checks. A model that leaks scores two. The worst scores one. That count is the model's certificate: it says how much of the model is checked, not how clever it is.

5 · What it means

Don't build on a leaking model, however good its graphs look. Spend money on measurements rather than on bigger computers: what is missing is data, not number-crunching. And say the honest sentence out loud — a projection for 2100 is an educated guess. What can be checked is whether the model is allowed to be believed.

6 · What this cannot do

We have not checked a single model against the actual future. Nobody has. Every projection you read is an extrapolation. The difference between a good one and a bad one is whether the model first passed the tests that would let you trust the extrapolation.